

Chapter 28 — Recovery and Continuity

Field	Value
Status	Revised after whole-book constitutional and cross-chapter audit
Version	0.2
Scope	Restore HAL after partial or catastrophic failure in an order that preserves constitutional identity, Owner authority, history, and verifiable continuity.
Constitutional basis	Decisions 22, 29, 35, 38, 42, 47, 50–51
Owner review	No unresolved Owner decision after whole-book audit

1. Purpose

Restore HAL after partial or catastrophic failure in an order that preserves constitutional identity, Owner authority, history, and verifiable continuity.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Recovery Manager	Recovery plans, orchestration, checkpoints, evidence, and disposition
Continuity Service	Constitution, identity, authority, audit, configuration, and ledger recovery order
Restore Verifier	Integrity, completeness, compatibility, identity continuity, and outcome tests
Recovery Presence	Minimal Owner interaction and status during degraded restoration

3. Core State and Records

Object	Required content
Recovery Plan	Scenario, priorities, sources, steps, authority, RPO/RTO, verification, and rollback.
Continuity Checkpoint	Constitution, Owner identity, authority, policy, audit, ledger versions, and signatures.
Restore Attempt	Source, target, versions, transformations, checks, result, and evidence.
Recovery Record	Failure, selected plan, timeline, restored state, gaps, uncertainty, approvals, and final status.

4. Runtime Workflow

1. Identify failure scope and preserve remaining evidence before making restorative changes.
2. Establish a trusted minimal environment and verify the Owner recovery channel.
3. Restore Constitution and Owner identity/authorization before other governance or capability.
4. Restore governance, audit, configuration, transaction/experience ledgers, and canonical knowledge in order.
5. Reconcile distributed state and rebuild derived indexes and caches from authoritative records.
6. Validate identity continuity, policy, trust, Treaties, transactions, capabilities, and limitations.
7. Resume through staged operation and issue a signed post-recovery Constitutional Mirror.

5. Interfaces and Contracts

Commands, queries, events, records, and evidence are typed, versioned, authenticated, authorized, correlated, and classified. Mutation is accepted only by the authoritative owner. Every external or reality-changing boundary carries identity, policy, authority, provenance, integrity, time, and transaction context.

6. Failure and Recovery

- Recovery source integrity uncertain: do not promote it; seek another independent copy or Owner-directed path.
- State versions conflict: preserve alternatives and reconstruct causality before selecting canonical state.
- Owner authentication unavailable: remain in minimal safe recovery without constitutional mutation.
- Restore passes technically but not constitutionally: keep production disabled until continuity checks pass.

7. Constitutional Guarantees

- Recovery restores who HAL is before restoring what HAL can do.
- Recovery evidence is explicit; absence of an error is not proof of continuity.
- Derived data is rebuilt only after canonical records are verified.
- Every material recovery remains reproducible, reviewable, and linked to the initiating incident.

8. Prohibited Behaviors

- No component may silently broaden or narrow constitutional authority, reinterpret Owner intent, erase dissent, or promote derived state to authority.
- No failure, degraded dependency, emergency, or optimization may silently change identity, policy, trust, authentication, evidence, or verification rules.
- No model, service, provider, node, Presence, credential, relationship, environment, or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records identity, correlation, authority/policy context, inputs, state versions, decisions, limitations, failures, recovery, and outcomes. Verification is risk-scaled, reproducible, and stored as durable evidence. Status and explanation derive from governed state rather than model assertion.

10. Source Alignment and Review

This chapter implements Decisions 22, 29, 35, 38, 42, 47, 50–51. Final source-alignment and constitutional disposition is recorded in the batch audit reports; embedded approval text is not treated as review evidence.

Whole-Book Audit Resolution

The following requirements supersede any ambiguous earlier wording in this chapter and are incorporated into this edition:

- A successor may receive a recovery constitutional lease only after independent state, identity, integrity, and quorum verification.
- The lease expires after no more than 24 hours. Every extension requires fresh, exact, Owner-specific authorization.
- Without extension, recovery continues only in Restricted or Safe Recovery mode and protected canonical mutation stops.
- Permanent reassignment of the Primary Constitutional Host requires the Owner Authorization Ceremony.